

Genetic report for longevity

Personality and Psychological Traits

Neuroticism and emotional stability, extroversion and sociability, taking risk and seeking novelty, well-being and satisfaction

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Personality traits are inherited in 40–50% in twin studies, but GWAS only explain a small part: neuroticism has hundreds of loci, risk taking and well-being dozens, and extraversion, openness, conscientiousness and pleasantness almost no confirmed locus. This report reads what is based on: neuroticism (two scores), taking risk (two scores that measure different things), social activity, satisfaction with friendships, happiness and well-being, and the loci of GWAS. Loneliness and sociability come from the Oxytocin and Social Behavior report; dopamine and serotonin genes of the profile (§15, §21), without judgment. Personality is not a disease: extreme traits have advantages and costs, and the favourable/unfavourable classification here refers only to the connection with health (neuroticism is associated with more illness and mortality).

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

16

variants

10

in the chip

5

imputed
($DR2 \geq 0.8$)

1

not found

11

Scores with
percentile

Overall reading. A social and stable to moderately emotional profile, with a particularity in risk taking. In neuroticism, the two polygenic scores are at the 73rd percentile and the 250 loci score in the 58: typical, slightly above the median. DCC in favor and XKR6 and MSRA typical. In 17q21, rs1052553 G/G indicates the **H2/H2** haplotype of CRHR1-MAPT inversion (about 6% of people); the CRHR1 alleles of more neuroticism and of more concern probably mark this haplotype, so they count as a single signal. In the relationships, the most favourable point: loneliness at the 8th percentile and sociability in 94 (Oxytocin Report and Social Behavior), weekly social activities in 68, satisfaction with friendships in 61; extraversion in 69 (only 4 loci). In taking risk, the scores are separated: consider someone who risks at the 91st percentile, but the factor of risk behaviors in four domains (conduction, finance, sex, tobacco/alcohol) in 13; tolerance to risk (277 loci) in 55. That is, more predisposition to see how someone who risks than risk behaviors; agrees with the PGS of having smoked at percentile 1.6. In well-being: less happiness in 39 and less well-being (194 loci) in 61, typical. DRD4 C/T, DRD2 without A1, COMT Val/Val, MAOA typical and OXTR G/G are left without judgment, as in the profile.

Favourable

- **Loneliness (PGS001091):** the 8th percentile (Oxytocin Report and Social Behavior)
- **Sociability (9 loci):** the 94th percentile (Oxytocin Report and Social Behavior)
- **DCC rs8084280 AA:** DCC: axonal guide; neuroticism and well-being.

Unfavourable

- **CRHR1 rs77804065 TT:** CRHR1: corticotropin-releasing hormone receptor (17q21).

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 10 chip genotypes, 0 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Neuroticism and emotional stability

Neuroticism (trend for negative emotions) is the most known trait and is associated with more depression, anxiety, cardiovascular disease and mortality. Strong loci: XKR6, MSRA, inversion 17q21 (CRHR1-MAPT) and DCC.

Ageing mechanisms: Intercellular communication alteration; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Neuroticism (950 183 variants)	Typical / intermediate	the 73rd percentile · z=0.56	Higher score : more neuroticism. PGS002213 (PGS Catalog; LDpred2 (bigsnp); Neuroticism score; PGP000263, PMID 34995502). Variants with genotype: 930594 of 950183 (covering of 96.9% coefficients).
Neuroticism (54 715 variants)	Typical / intermediate	the 74th percentile · z=0.58	Higher score : more neuroticism. PGS001996 (PGS Catalog; Penalized regression (bigstatsr); Neuroticism score; PGP000263, PMID 34995502). Variants with genotype: 52768 of 54715 (covering of 96.4% coefficients).
Neuroticism (GWAS loci)	Typical / intermediate	the 58th percentile · z=0.18	Higher score : more neuroticism. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 250 of 252 loci with genotype. Studies: Baselmans BML (Neuroticism); Cai N (Neuroticism); Luciano M (Neuroticism); Nagel M (Neuroticism); Okbay A (Neuroticism); Smith DJ (Neuroticism); Moor MH (Neuroticism). Variants with genotype: 250 of 252 (covering of 99.2% coefficients).

Favourable

- DCC rs8084280 AA

Unfavourable

- CRHR1 rs77804065 TT

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DCC	rs8084280 18:53200379 · A/T	AA	imputation DR2=1.00	T (0; 48%)	moderate	DCC: axonal guide; neuroticism and well-being. Meaning: GWAS Catalog: Neuroticism (Baselmans BML 2019; $p=3e-19$); T ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CRHR1	rs77804065 17:45733530 · C/A/G/T	TT	imputation DR2=0.99	T (2; 25%)	moderate	CRHR1: corticotropin-releasing hormone receptor (17q21). Probably marks the haplotype H2 of inversion 17q21 (see rs1052553). Meaning: GWAS Catalog: Neuroticism (Nagel M 2018; $p=1e-31$); T ↑ allele

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
XKR6	rs6981523 8:11204283 · C/A/T	CT	chip	T (1; 50%)	moderate	XKR6: one of the strongest loci of neuroticism (8p23 chromosome). Meaning: GWAS Catalog: Neuroticism (Baselmans BML 2019; $p=7e-32$); T ↑ allele
MSRA	rs6982308 8:10336262 · C/G	CG	chip	G (1; 55%)	moderate	MSRA: methionine sulfoxide reductase (repair of oxidized proteins); locus of neuroticism. Meaning: GWAS Catalog: Neuroticism (Baselmans BML 2019; $p=6e-31$); G ↑ allele

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MAPT (17q21 inversion)	rs1052553 17:45996523 · A/G	GG	chip	—	moderate	H1/H2 haplotype marker of 17q21 inversion (CRHR1, MAPT, KANSL1): H2 is associated with less neuroticism in some studies and changes the risk of neurodegenerative diseases. Information. G/G = H2/H2 (about 6% of people in the reference population). The CRHR1 alleles of more neuroticism (here) and of more concern (anxiety report) have the frequency of H2 and probably mark the same haplotype.

Extraversion, sociability and relationships

Extroversion has only 4 loci confirmed; loneliness and sociability come from the Oxytocin and Social Behavior report (OXTR rs53576 G/G). Chronic loneliness is associated with more inflammation and mortality.

Ageing mechanisms: Intercellular communication alteration; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Social or leisure activities weekly (2911 variants)	No judgement (informative)	the 68th percentile · z=0.50	Information. PGS001018 (PGS Catalog; snpnet; Social attention/leisure activities (attend any of the followings in once a week or more often: sports club, gym, pub, social club, religious group, adult edition class, or other group activity); GPG000244, PMID 35324888). Variants with genotype: 2555 of 2911 (covering of 88.4% coefficients).
Satisfaction with friendships (659 variants)	No judgement (informative)	the 61st percentile · z=0.27	UK Biobank scale with meaning not indicated in the catalog; informative. PGS001398 (PGS Catalog; snpnet; Friendship satisfaction; GPG000244, PMID 35324888). Variants with genotype: 575 of 659 (covering of 88.2% coefficients).
Extraversion (4 loci)	No judgement (informative)	the 69th percentile · z=0.44	Higher score = more extroversion. Information; only 4 loci. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 4 of 4 loci with genotype. Studies: Lo MT (Extraversion). Variants with genotype: 4 of 4 (covering of 100.0% coefficients).
Loneliness (PGS001091)	Favourable	the 8th percentile	Higher score : more loneliness. Source: Oxytocin report and Social Behavior.
Sociability (9 loci)	Favourable	the 94th percentile	Higher score : more sociability. Source: Oxytocin report and Social Behavior.

Favourable

- Loneliness (PGS001091) (the 8th percentile)
- Sociability (9 loci) (the 94th percentile)

Unfavourable

None.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
OXTR	rs53576 3:8762685 · A/G/T	GG	chip	—	weak	OXTR rs53576 G/G was associated in small studies with more empathy and sociability; it did not replicate in large ones. Detailed in the Oxytocin and Social Behavior report. G/G (coincidence with Oxytocin report).
PCDH15	rs6481128 10:54901602 · G/A	AA	chip	G (0; 45%)	weak	PCDH15: protocaderin; extroversion. Informative: 0 copy(s) of the G allele, the most extroversion in GWAS. Meaning: GWAS Catalog: Extraversion (Lo MT 2016; p=5e-10); allele G ↑

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
WSCD2	rs1426371 — · —	—	SNP not found in the data	—	weak	WSCD2: the strongest locus of extroversion (only confirmed 4 loci). No usable genotype.

Risk taking and novelty search

CADM2 is the strongest locus of risk tolerance; DRD4, DRD2 and COMT are the classic candidates of novelty search (without judgment, as in §21). The two risk scores measure different things: a single question "is considered someone at risk?", the other one is a common factor of four domains (conduction, finance, sex, tobacco/alcohol).

Ageing mechanisms: Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
It is considered someone who risks (4297 variants)	No judgement (informative)	the 91st percentile · z=1.40	Higher score = more risk taking. Information. PGS001049 (PGS Catalog; snpnet; Risk taking behavior; PGP000244, PMID 35324888). Variants with genotype: 3782 of 4297 (covering of 88.7% coefficients).
Risk taking in 4 domains (1 110 737 variants)	No judgement (informative)	the 13th percentile · z=-1.19	Higher score : more risk in driving, finance, sex and tobacco/alcohol. Information. PGS000205 (PGS Catalog; PRS-CS; Risk-taking trend (4-domain main component model); PGP000084, PMID 32555147). Variants with genotype: 1078541 of 1110737 (covering of 96.7% coefficients).
Risk tolerance (GWAS loci)	No judgement (informative)	the 55th percentile · z=0.03	Higher score : more tolerance to risk. Information. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 277 of 282 loci with genotype. Studies: Clifton EAD (Self-reported risk-taking behavior); Karlsson Linner R (General risk tolerance (MTAG)); Karlsson Linner R (Risk-taking tendency (4-domain principal component model)); Strawbridge RJ (Self-reported risk-taking behavior). Variants with genotype: 277 of 282 (covering of 98.2% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (6)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DRD4	rs1800955 11:636784 · T/C/G	CT	chip	—	weak	DRD4 -521C>T: T allele lowers expression; weak association with novelty. The VNTR of the exon 3 (7R) is not read by chip.
DRD2/ANKK1	rs1800497 11:113400106 · G/A	GG	chip	—	weak	DRD2/ANKK1 Taq1A: the allele A1 is associated with less D2 receptors; G/G = without A1. G/G = without the A1 allele.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Val/Val is associated on average with less avoidance of danger and more search for novelty (small effect). G/G = Val/Val.
MAOA	rs6323 X:43731789 · G/T	T	chip	—	weak	MAOA (chromosome X, 1 copy): typical activity in §21. The VNTR of the promoter (MAOA-L/H), the classical marker, is not read by chip. T (1 copy, X): "type" in §21.
CADM2	rs6783138 3:85550770 · C/A/G/T	GC	imputation DR2=1.00	C (1; 37%)	moderate	CADM2: neuronal adhesion; the strongest locus of risk tolerance (also of concern and body composition). Information: 1 copy(s) of the C allele, the one with more risk tolerance in GWAS. Meaning: GWAS Catalog: General risk tolerance (MTAG) (Karlsson Linner R 2019; p=1e-90); C ↑ allele
FOXP2	rs727644 7:114469294 · A/G/T	AG	imputation DR2=0.99	G (1; 52%)	moderate	FOXP2: risk tolerance and ADHD. Information: 1 copy(s) of the G allele, the one with more risk tolerance in GWAS. Meaning: GWAS

Well-being, happiness and satisfaction

The subjective well-being shares (with opposite signal) with depression and neuroticism. UK Biobank happiness score and well-being locus and life satisfaction.

Ageing mechanisms: Chronic Inflammation; Endocrine dysregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Less general happiness (10 628 variants)	Typical / intermediate	the 39th percentile · z=-0.27	Higher score = less happy (UK Biobank question with inverted scale). PGS001936 (PGS Catalog; Penalized regression (bigstatsr); General happiness; PGP000263, PMID 34995502). Variants with genotype: 10306 of 10628 (covering of 96.5% coefficients).
Less subjective well-being (GWAS loci)	Typical / intermediate	the 61st percentile · z=0.26	Higher score : less well-being. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 194 of 197 loci with genotype. Studies: Baselmans BML (Life satisfaction); Baselmans BML (Well-being spectrum) Kim S (Subjective well-being); Okbay A (Life satisfaction); Okbay A (Subjective well-being); Turley P (Subjective well-being). Variants with genotype: 194 of 197 (covering of 92.3% coefficients).

Favourable

None.

Unfavourable

None.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GRM5	rs597462 11:88955065 · T/A/C	TC	imputation DR2=1.00	C (1; 51%)	moderate	GRM5: metabotropic glutamate receptor; well-being. Meaning: GWAS Catalog: Well-being spectrum (multivariate analysis) (Baselmans BML 2019; p=2e-19); allele C ↓
CELF4	rs11082011 18:37565159 · C/G/T	CT	chip	T (1; 66%)	moderate	CELF4: regulation of neuronal RNA; well-being. Meaning: GWAS Catalog: Well-being spectrum (multivariate analysis) (Baselmans BML 2019; p=7e-19); T ↓ allele

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	CRHR1 TT	DCC AA; Loneliness (PGS001091) (the 8th percentile); Sociability (9 loci) (the 94th percentile)	Neuroticism (950 183 variants) (the 73rd percentile); Neuroticism (54 715 variants) (the 74th percentile); Neuroticism (GWAS loci) (the 58th percentile); XKR6 CT; MSRA CG
Chronic Inflammation	CRHR1 TT	DCC AA; Loneliness (PGS001091) (the 8th percentile); Sociability (9 loci) (the 94th percentile)	Neuroticism (950 183 variants) (the 73rd percentile); Neuroticism (54 715 variants) (the 74th percentile); Neuroticism (GWAS loci) (the 58th percentile); XKR6 CT; MSRA CG; Less general happiness (10 628 variants) (the 39th percentile); Less subjective well-being (GWAS loci) (the 61st percentile); GRM5 TC
Endocrine dysregulation	—	—	Less general happiness (10 628 variants) (the 39th percentile); Less subjective well-being (GWAS loci) (the 61st percentile); GRM5 TC; CELF4 CT

Practical part

Ideas for personality and well-being, crossed with the genetic plan v2 and current medication. The current stack remains as it is; everything is suggestion (not applied). Personality does not "correct"; the objective is to use strengths (relationships) and put rules where genetic pulls (risk, rumination).

1. Investing in relationships as a pillar of longevity

Genetics: Solitude p8; Sociability p94

The close relationships are associated with less mortality, with an effect comparable to that of tobacco. Maintaining regular meetings and group activities (sports, clubs) takes advantage of a favourable predisposition.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. One change at a time in the stack

Genetics: It is considered someone who risks p91

With a tendency to risk decisions, changing one thing at a time (supplement, drug, dose), with registration and with the doctor, allows us to know what works and catch adverse effects early.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Routines against rumination

Genetics: Neuroticism p73; 17q21 H2/H2

Regular exercise, fixed sleep and 10 minutes per day of writing or mindfulness decrease neuroticism in trials (small to moderate effect).

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

4. Activities with meaning

Genetics: Typical well-being (p39–p61)

Volunteering, learning new things and long-term goals are associated with more well-being and less mortality in cohort studies.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Big decisions after bad nights. *suggestion (not applied)*
- Add several changes in the stack at the same time. *suggestion (not applied)*
- Prolonged isolation (e.g. weeks of remote work without social contact). *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. VNTR of DRD4 and MAOA do not read by chip.
- Openness, conscientiousness and kindness are not confirmed enough for a score; they are not evaluated.
- The UK Biobank scores come from unique questions; the self-declared risk taking and the factor of four domains are different constructs and may disagree.
- Personality is not a disease; favourable/unfavourable refers only to the connection with health.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS002213 (PGS Catalog; LDpred2 (bigsnpr); Neuroticism score; PGP000263, PMID 34995502 · PGS001996 (PGS Catalog; Penalized regression (bigstatsr); Neuroticism score; PGP000263, PMID 34995502 · PGS001018 (PGS Catalog; snpnet; Social attention/leisure activities (attend any of the followings in once a week or more often: sports club, gym, pub, social club, religious group, adult edition class, or other group activity); PGP000244, PMID 35324888 · PGS001398 (PGS Catalog; snpnet; Friendship satisfaction; PGP000244, PMID 35324888 · PGS001049 (PGS Catalog; snpnet; Risk taking behavior; PGP000244, PMID 35324888 · PGS000205 (PGS Catalog; PRS-CS; Risk-taking trend (4-domain main component model); PGP000084, PMID 32555147 · PGS001936 (PGS Catalog; Penalized regression (bigstatsr); General happiness; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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- Tanigawa Y et al. Significant sparse polygenic risk scores across 813 traits in UK Biobank. *PLoS Genet* 2022 (PGS001018, PGS001049, PGS001398).
- Privé F et al. Portability of 245 polygenic scores. *Am J Hum Genet* 2022 (PGS001936, PGS001996, PGS002213).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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